Asssignment 4

Astrophysics 509

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Question 1

the dispersion relation for electromagnetic waves is given by

$$\omega^2 = \frac{c^2 k^2}{\epsilon} \tag{1.1}$$

Where ϵ is the dielectric constant of the medium through which the wave propagates. For many astrophysical plasmas, the permittivity can be approximated as $\epsilon \approx 1 - \omega_p^2/\omega^2$, where

$$\omega_p^2 = 4\pi n_e \frac{e^2}{m_e} \tag{1.2}$$

is the plasma frequency. Work out the group velocity $v_{group} = \frac{\partial \omega}{\partial k}$ and the travel time.

$$t_{travel} = \frac{d}{c} + \frac{2\pi e^2}{m_e \omega^2} DM \tag{1.3}$$

Where d/c is the travel time at very high frequencies, and DM is the dispersion measure defined as:

$$DM = \int_0^d n_e \, \mathrm{d}s \tag{1.4}$$

Equation 5.73 is missing factor of 1/c on the RHS

Let's input ϵ into the expression for the dispersion relation:

$$\omega^2 = \frac{c^2 k^2}{1 - \omega_p^2/\omega^2} \tag{1.5}$$

Let's put this in terms of k to take advantage of this random ahh $1 - \omega_p^2/\omega^2$

$$k = \frac{\omega}{c}\sqrt{1 - \omega_p^2/\omega^2} \tag{1.6}$$

The partial w.r.t ω is then:

$$\frac{\partial k}{\partial \omega} = \frac{1}{c}\sqrt{1 - \omega_p^2/\omega^2} + \frac{\omega}{2c}\left(1 - \omega_p^2/\omega^2\right)^{-1/2}\left(\frac{-2\omega_p^2}{\omega^3}\right) \tag{1.7}$$

$$= \frac{1}{c}\left(1 - \omega_p^2\omega^2\right)^{-1/2} \tag{1.8}$$

The group velocity given by the following partial is then:

$$v_{group} = \left(\frac{\partial k}{\partial \omega}\right)^{-1} = c\sqrt{1 - \omega_p^2/\omega^2} \tag{1.9}$$

The travel time is then:

$$t_{travel} = \int_0^d \frac{1}{v_{group}} \, \mathrm{d}s \tag{1.10}$$

$$= \int_0^d \frac{1}{c\sqrt{1-\omega_p^2/\omega^2}} \, \mathrm{d}s \tag{1.11}$$

$$= \int_0^d \frac{1}{c}\left(1-\omega_p^2/\omega^2\right)^{-1/2} \mathrm{d}s \tag{1.12}$$

$$= \int_0^d \frac{1}{c}\left(1-\frac{4\pi n_e e^2}{m_e\omega^2}\right)^{-1/2} \mathrm{d}s \tag{1.13}$$

$$= \int_0^d \frac{1}{c}\left(1+\frac{2\pi n_e e^2}{m_e\omega^2}\right) \mathrm{d}s \tag{1.14}$$

Where we used the binomial expansion to get rid of the square root. The itegral is then:

$$\frac{d}{c} + \frac{2\pi e^2}{c m_e \omega^2} \int_0^d n_e \, \mathrm{d}s \tag{1.15}$$

Where we finally have:

$$t_{travel} = \frac{d}{c} + \frac{2\pi e^2}{c m_e \omega^2} DM \tag{1.16}$$

Question 2

Pulsar Energetics

Question 2.a

Estimate the magnetic field of a pulsar (1 solar mass, radius of 10km). Assume it forms from a star of 1 solar mass and 1 solar radius with dipole magnetic field with polar surface field strength of 200 Gauss. Assume flux conservation.

The flux conservation is key to this estimate. Since the area goes as radius squared, we get::

$$\frac{B_p(1\,\mathrm{R}_\odot)}{B_p(10\,\mathrm{km})} = \frac{(10\,\mathrm{km})^2}{(1\,\mathrm{R}_\odot)^2} \tag{2.1}$$

$$B_p(10\,\mathrm{km}) = \frac{(1\,\mathrm{R}_\odot)^2 B_p(1\,\mathrm{R}_\odot)}{(10\,\mathrm{km})^2} = \frac{(6.960\times10^6\,\mathrm{km})^2(200\,\mathrm{G})}{(10\,\mathrm{km})^2} = 9.69\times10^{13}\,\mathrm{G} \tag{2.2}$$

Question 2.b

Estimate the kick velocity of a neutron star. Assume a uniform density $M = 1.4\,\mathrm{M}_\odot$, $R = 10\,\mathrm{km}$ neutron star and calculate the energy release, assuming an initial uniform density core with radius of 2000 km. Assume 99% of the collapse energy is radiated as neutrinos; and assume the momentum asymmetry of the emitted neutrinos is 1%.

The typical energy released in neutrinos for a new neutron star is 1×10^{53} erg. If we assume that 1% of this is then asymmetric, we can say that this energy directly kicks the neutron star back. This then

$$E_{kick} = 1 \times 10^{51}\,\mathrm{erg} = (\gamma - 1)mc^2 \tag{2.3}$$

Where we get the velocity as:

$$v = c\sqrt{1 - \left(\frac{mc^2}{E_{kick} + mc^2}\right)^2} = c\sqrt{1 - \left(\frac{(1.4 \cdot 1.989 \times 10^{33}\,\mathrm{g})c^2}{1 \times 10^{51}\,\mathrm{erg} + (1.4 \cdot 1.989 \times 10^{33}\,\mathrm{g})c^2}\right)^2} \tag{2.4}$$

When $c = 2.998 \times 10^{10}\,\mathrm{cm/s}$, we get:

$$v = 8.44 \times 10^{8}\,\mathrm{cm/s} = 8.44 \times 10^{3}\,\mathrm{km/s} \tag{2.5}$$

Question 2.c

Calculate I for a neutron star in terms of M and R, assuming uniform density (show your calculation of I, using the definition of moment of inertia of an extended object). Give the numerical value for a $M = 1.4\,\mathrm{M}_\odot$, $R = 12\,\mathrm{km}$ neutron star.

The definition goes as:

$$I = \int r^2\,\mathrm{d}m \tag{2.6}$$

where r is the distance away from the axis of rotation. Since we can define this axis to be the z-axis, and we take the coordinate to be:

$$z = r\sin\theta \tag{2.7}$$

where r is the r in spherical coordinates. Turning this integral into a volume integral where $\mathrm{d}m = \rho\,\mathrm{d}V = \rho r^2 \sin\theta\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}\phi$.

$$I = \int_0^R \int_0^\pi \int_0^{2\pi} (r\sin\theta)^2 \rho r^2 \sin\theta\,\mathrm{d}\phi\,\mathrm{d}\theta\,\mathrm{d}r \tag{2.8}$$

$$I = \rho \int_0^R r^4\,\mathrm{d}r \int_0^\pi \sin^3\theta\,\mathrm{d}\theta \int_0^{2\pi} \mathrm{d}\phi \tag{2.9}$$

$$I = \rho\left(\frac{R^5}{5}\right)\left(\frac{4}{3}\right)(2\pi) \tag{2.10}$$

Since we have uniform density, we get with the density:

$$I = \left(\frac{M}{(4/3)\pi R^3}\right)\left(\frac{8\pi R^5}{15}\right) = \frac{2MR^2}{5} \tag{2.11}$$

Inserting numbers we get:

$$I = \frac{2MR^2}{5} = \frac{2(1.4 \cdot 1.989 \times 10^{33}\,\mathrm{g})(12 \cdot 10^5 \mathrm{cm})^2}{5} = 1.60 \times 10^{45}\,\mathrm{g\ cm}^2 \tag{2.12}$$

Question 3

Magnetic Dipole

Question 3.a

Derive the formula for the energy loss rate and rate of change of angular frequency and of period, for a rotating magnetic dipole with tilt angle α between the magnetic and rotation axes. Start with the Larmor formula and show all steps required for the calculation. Write the solution in terms of the polar magnetic field, B_p.

We have Larmor formula as:

$$\frac{\mathrm{d}W}{\mathrm{d}t} = -\frac{2}{2c^3}\left|\ddot{\vec{d}}\right|^2 \tag{3.1}$$

For a magnetic dipole we get the equivalent:

$$\frac{\mathrm{d}W}{\mathrm{d}t} = -\frac{2}{2c^3}\left|\ddot{\vec{m}}\right|^2 \tag{3.2}$$

Where m is the magnetic dipole moment. We then have $\vec{m} = m_0\hat{e}_m$, where $m_0 = B_pR^3/2$ and $\hat{e}_m = (\sin\alpha\cos\omega t, \sin\alpha\sin\omega t, \cos\alpha)$. The time derivatives are then:

$$\frac{\mathrm{d}\hat{e}_m}{\mathrm{d}t} = \omega(-\sin\alpha\sin\omega t, \sin\alpha\cos\omega t, 0) \tag{3.3}$$

$$\frac{\mathrm{d}^2\hat{e}_m}{\mathrm{d}t^2} = \omega^2(-\sin\alpha\cos\omega t, -\sin\alpha\sin\omega t, 0) \tag{3.4}$$

Taking the squared absolute value gives:

$$\left(\sqrt{\omega^4\sin^2\alpha\sin^2\omega t + \omega^4\sin^2\alpha\cos^2\omega t}\right)^2 = \omega^4\sin^2\alpha(\sin^2\omega t + \cos^2\omega t) \tag{3.5}$$

The extra factor at the end is identically 1 so we finally have:

$$\left|\ddot{\vec{m}}\right|^2 = m_0^2\omega^4\sin^2\alpha \tag{3.6}$$

So we finally have:

$$\boxed{\frac{\mathrm{d}W}{\mathrm{d}t} = -\frac{B_p^2R^6}{6c^3}\omega^4\sin^2\alpha} \tag{3.7}$$

Relating this to the change in angular frequency comes from the rotational energy:

$$E_{rot} = \frac{1}{2} I \omega^2 \tag{3.8}$$

$$\frac{\mathrm{d}E_{rot}}{\mathrm{d}t} = I\omega\dot{\omega} \tag{3.9}$$

Omg we can say that this is identical to the energy loss rate, and so we can equate them:

$$I\omega\dot{\omega} = -\frac{B_p^2 R^6}{6c^3}\omega^4 \sin^2\alpha \tag{3.10}$$

$$\dot{\omega} = -\frac{B_p^2 R^6 \omega^3 \sin^2\alpha}{6c^3 I} \tag{3.11}$$

This quantity will always be less that zero, which shows that pulsars slow down. The period slow down is then just:

$$\dot{P} = -\frac{2\pi}{\omega^2}\dot{\omega} = \frac{\pi B_p^2 R^6 \omega^3 \sin^2\alpha}{3c^3 I} \tag{3.12}$$

Question 4

Assume that Eq: 6.48

$$\frac{\mathrm{d}a}{\mathrm{d}t} = -\frac{64G^3 m_1 m_2 M}{5a^3c^5}\frac{(1 + 73e^2/24 + 37e^4/96)}{(1-e^2)^{5/2}} \tag{4.1}$$

remains valid until the stars come into contact. SHow that the inspiral time from the current orbital separation a_0 for a circular binary orbit is given by:

$$\tau_{insp} = \frac{5}{256}\frac{c^5}{G^3}\frac{a_0^4}{M^2\mu} \tag{4.2}$$

Where M is the total and μ the reduced mass.

We get a reduced equation when we consider zero eccentricity:

$$\frac{\mathrm{d}a}{\mathrm{d}t} = -\frac{64G^3 m_1 m_2 M}{5a^3c^5} \tag{4.3}$$

Let's absorb $m_1 m_2 = \mu M$:

$$\frac{\mathrm{d}a}{\mathrm{d}t} = -\frac{64G^3 \mu M^2}{5a^3c^5} \tag{4.4}$$

We can then treat this as a separable equation to recover the inspiral time (from when we go from $a_0 \to 0$):

$$a^3\,\mathrm{d}a = -\frac{64G^3\mu M^2}{5c^5}\,\mathrm{d}t \tag{4.5}$$

$$\int_{a_0}^{0} a^3\,\mathrm{d}a = \int_0^{\tau_{insp}} -\frac{64G^3\mu M^2}{5c^5}\,\mathrm{d}t \tag{4.6}$$

$$-\frac{a_0^4}{4} = -\frac{64G^3\mu M^2}{5c^5}\tau_{insp} \tag{4.7}$$

Where we finally have:

$$\tau_{insp} = \frac{5}{256}\frac{c^5}{G^3}\frac{a_0^4}{M^2\mu} \tag{4.8}$$

Question 4.a

Use equation 6.48 of the text to derive the inspiral time from initial orbital separation a_0 to final separation a_f when the two stars come into contact. Assume zero eccentricity.

$$\tau_{insp} = \frac{5}{256}\frac{c^5}{G^3}\frac{a_0^4 - a_f^4}{M^2\mu} \tag{4.9}$$

Where we simply had a non zero bound on our in our integration of the separation.

Question 4.b

A system of two neutron stars, each of mass 1.4 Msun and radius 10 km, has circular orbit with initial orbital period of 60 s. Find the inspiral time.

We have that $M = m_1 + m_2 = 2m_1$, which means that $\mu = m_1/2$. From the period, we know that Kelper's third gives:

$$T^2 = \frac{4\pi^2}{GM}a^3 \tag{4.10}$$

So our separation is then:

$$a = \left(\frac{GMT^2}{4\pi^2}\right)^{1/3} \tag{4.11}$$

The insp time is then:

$$\tau_{insp} = \frac{5}{256}\frac{c^5}{G^3}\frac{a^4}{M^2\mu} \tag{4.12}$$

Where we neglect the final separation of 20 km

$$\tau_{insp} = \frac{5}{256}\frac{(2.998\times10^{10}\,\mathrm{cm/s})^5}{(6.673\times10^{-8}\,\mathrm{dyne\ cm^2\ g^{-2}})^3}\frac{1}{2(1.4\cdot1.989\times10^{33}\,\mathrm{g})^3}\left(\frac{6.673\times10^{-8}\,\mathrm{dyne\ cm^2\ g^{-2}}(2\cdot1.4\cdot1.989\times10^{33}\,\mathrm{g})(60s)^2}{4\pi^2}\right)^{4/} \tag{4.13}$$

Aight lets calculate this:

$$\tau_{insp} = 4.04\times10^9\,\mathrm{s} \tag{4.14}$$

Question 5

Binary Evolution

Question 5.a

A 5 solar mass main sequence star is in circular orbit with a 5 solar mass compact object (black hole). An isolated 5 solar mass star has its radius increase from 2.7 to 7 $R_\odot$ during the main sequence phase. During the red giant phase it expands from 7 to 100 $R_\odot$, and during the AGB phase its radius increases from 100 to 1000 $R_\odot$.
What orbital period ranges will result in Roche Lobe overflow (onto the black hole) during each of the three phases? (main sequence, red giant, and AGB). The radius of a sphere with the same volume as the Roche Lobe around mass M1 is given by: $R_L/a = 0.49q^{2/3}/(0.60q^{2/3} + \ln(1 + q^{1/3}))$

References